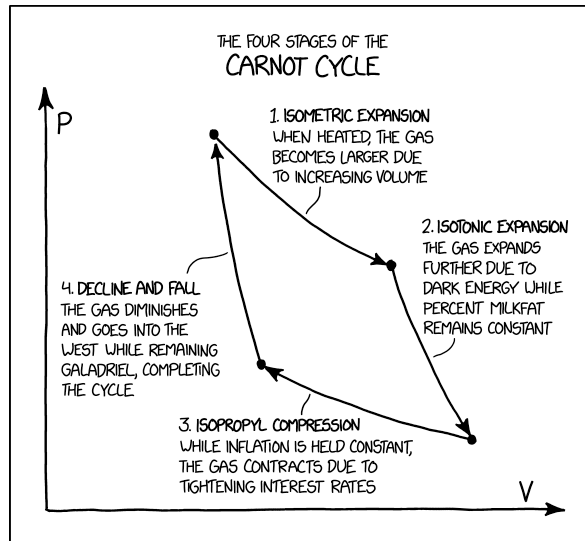




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(1)

Pikes Peak 2010

Pikes Peak Time Attack Class – Porsche 911 GT3 Cup class record – last before course fully paved...



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(2)

<https://youtu.be/1CxVLRjNVvU>

ME416

Ground Vehicle Dynamics

Performance –
Power and Transmission

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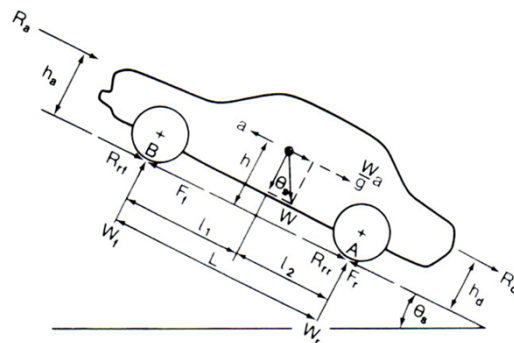
[3]

Driving Resistances

- Previously we identified the total driving resistance and therefore demand for tractive force, F_T , of the vehicle

$$F_T = F_w + F_d + F_g + F_a$$

$$F_T = \mu_T W \cos \theta + C_D \frac{1}{2} \rho V^2 A_f + W \sin \theta + (m_{unladen} * \gamma + m_{add}) * \ddot{x}$$



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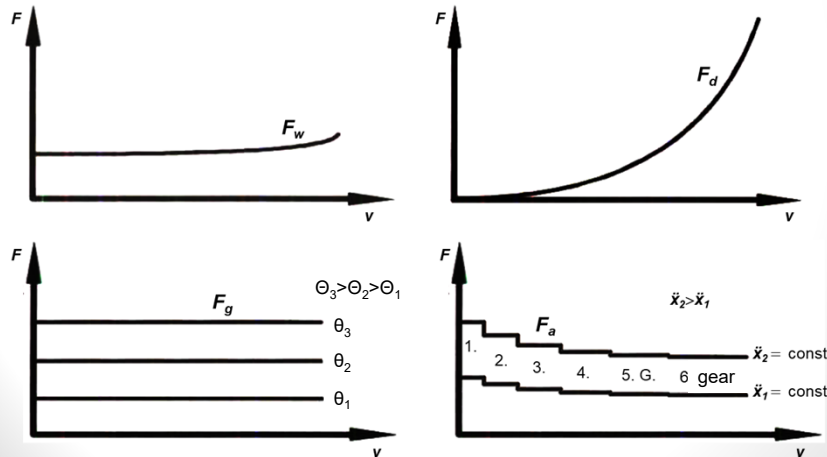
[4]

Driving Resistances (2)

- Previously we identified the total driving resistance and therefore demand for tractive force, F_T , of the vehicle

$$F_T = F_w + F_d + F_g + F_a$$

$$F_T = \mu_T W \cos \theta + C_D \frac{1}{2} \rho V^2 A_f + W \sin \theta + (m_{untladen} * \gamma + m_{add}) * \ddot{x}$$

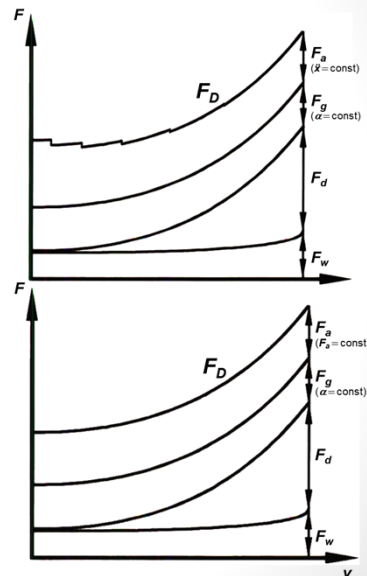


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[5]

Driving Resistances (3)

- Top
 - Tractive force demand for constant car acceleration, $\ddot{x}$
 - Note the reductions in F_a as the effective mass term reduces as each (finite) gear is selected
 - This form is used to determine the required torque
- Bottom
 - Tractive force demand for constant acceleration force, F_a
 - This form is used to determine the force balance at the tire/ground

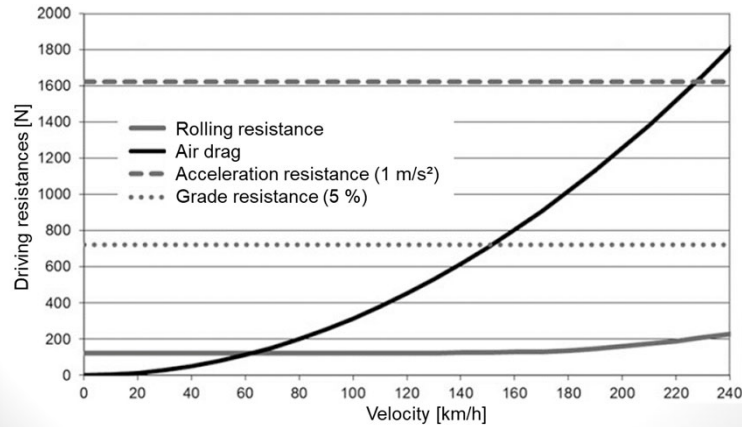


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[6]

Driving Resistances (4)

- Mercedes B-class, 2013
 - 1500kg, $C_D=0.26$, $A_f=2.42\text{m}^2$, $\mu_r=0.0085$
 - Air drag exceeds wheel drag at 60kph (37mph)
 - Air drag at 190kph (117mph) ~10x as at 60kph (37mph)
 - Gradient drag of 5% slope same as air drag at 150kph (92mph)
 - Acceleration at 1m/s^2 (3ft/s^2 , $0.1g$) equals air drag at 230kph (142mph)!



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[7]

Ideal Motor

- Now, knowing our loads, ideally we would like to have an engine/motor that has a constant power vs. speed
 - $P=\text{constant}$

$$P = 2\pi nT$$

$$T = \frac{P}{2\pi n}$$

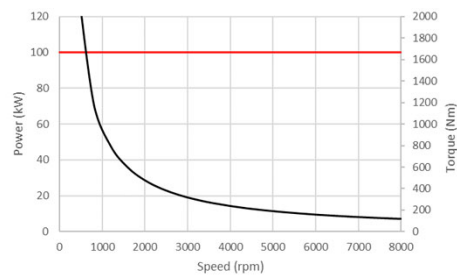
Where,

P =power

T =torque

n =motor/engine rotational speed (rev/s)

- For example, for $P=100\text{kW}$

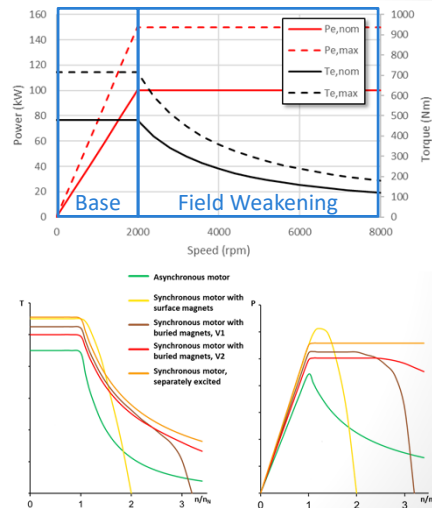


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[8]

Ideal Electric Motor

- An ideal electric motor will have
 - Nominal power and torque (solid lines)
 - Max power and torque (dashed lines)
 - Short term use only
- Ranges of operation can be defined
 - Base speed range
 - Linear rise of power with constant torque
 - Field weakening range
 - Constant power with decreasing torque due to weakening of field intensity up to n_{max}
- Other limiting factors
 - Temperature/cooling
 - Mechanical strength/durability
 - Wire diameter (current capacity)
- Some in use and possible motor constructions torque and power curves are presented

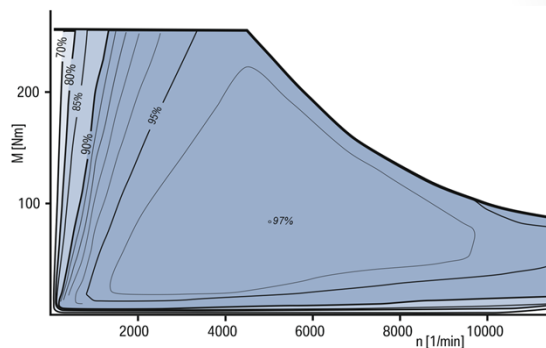


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9

Electric Motor Efficiency

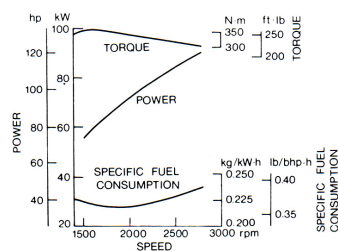
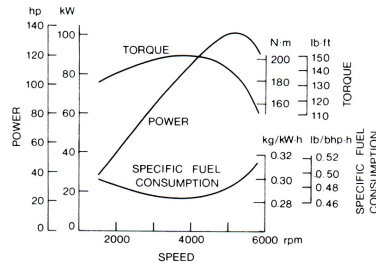
- To the right is an example of an electric vehicle motor efficiency plot (BMW i3)
 - High efficiency (>97%) over a large range
 - At low speeds and low torques the efficiency falls off quickly
 - Note that this is also a liquid cooled motor



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10

Power Characteristics, IC Engine



- As previously stated, ideally we would like to have an engine/motor that has a constant output vs. speed
 - Electric motors can be close to the ideal
 - IC engines are not
- From the typical plots at right (petrol top – diesel bottom) we see that the IC engine does not produce significant torque at very low speeds, i.e. near zero rpm
- For good performance from standstill and low speeds as well as sufficiently high top speeds we will need to provide for a transmission with multiple or variable ratios
 - We also need a little more background to discuss efficiency of IC engines

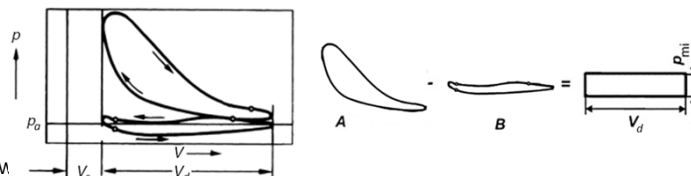
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[11]

Power and MEP

For all intents and purposes, all current IC engines are reciprocating piston engines and we will focus on those

- Gas forces provide the indicated work, W_i , which is the integral of the piston pressure, p , with respect to the piston travel, s_p , multiplied by the piston area A_p .



- W_i
- A rectangle of the same area has the displacement V_d as length and the indicated mean effective pressure p_{mi} as height.
- Indicated work equals indicated power (for a 4-stroke engine where c =number of cylinders and n = engine speed in rev/s)

$$P_i = p_{mi} A_p c \frac{n}{2}$$

- Indicated power minus friction power yields effective power

$$P_e = P_i - P_f$$

- Brake mean effective pressure (bmep), P_{me}

$$P_{me} = \frac{P_e}{A_p c \frac{n}{2}} = \frac{P_e}{V_d \frac{n}{2}} = \frac{T_e 4\pi}{V_d} = \frac{P_e}{V_d \frac{n}{2}}$$

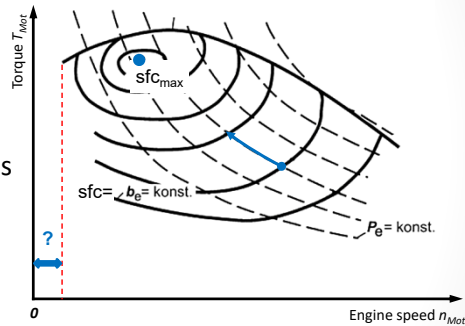
- Therefore the engine torque, T_e , and engine power, P_e , can be obtained from the bmep, P_{me} , and the engine displacement, V_d

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[12]

Engine Characteristic Map

- Upper boundary is full load or performance curve
- Lines of constant specific fuel consumption
 $sfc(b_e) = \text{const}(g/kWh)$
- Optimum operating point is $sfc_{\max}$ with highest efficiency
- Specific consumption depends on P_e required and engine speed
- Starting gap: $n_{\min} > 0$



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[13]

Notable Points in Characteristic Map

- Fuel Consumption

$$\frac{P_e(sfc)}{V_{veh}\rho_{fuel}} \left[\frac{l}{100km} \right] \text{ or } P_e(sfc) \left[\frac{g}{hr} \right]$$

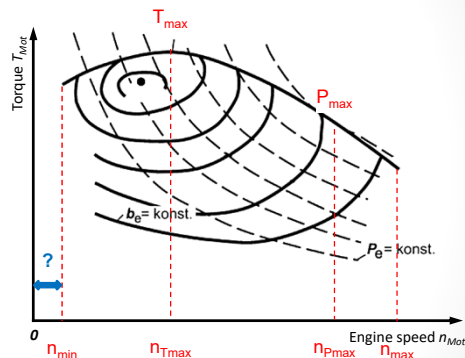
P_e =power (required) of engine (kW)

V_{veh} = velocity of vehicle (m/s)

sfc =specific fuel consumption (g/kWh)

ρ_{fuel} =fuel density (kg/m³)

Careful with units. Customary equivalents is mpg and lb/hr.



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[14]

Some IC Engine Background

- A gasoline engine is capable of operating on air to fuel ratios of 8:1 to 20:1
 - The optimum range is generally (depending on gasoline composition and what 'optimum' is desired) is generally 14:1 to 15:1
 - A air to fuel ratio of 14.5:1 is generally assumed as stoichiometric
- We can define an air factor,

$$\lambda = \frac{\lambda_{\text{actual}}}{\lambda_{\text{stoichiometric}}}$$

where,

$\lambda > 1$ is lean or weak, i.e. more air than fuel

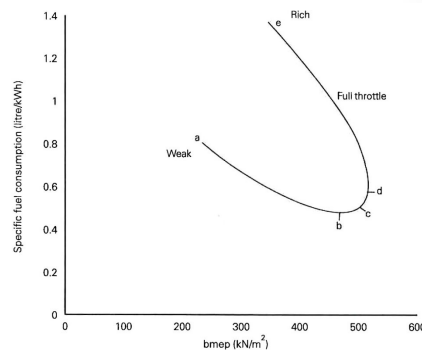
$\lambda < 1$ is rich, i.e. more fuel than air

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[15]

Some IC Engine Background (2)

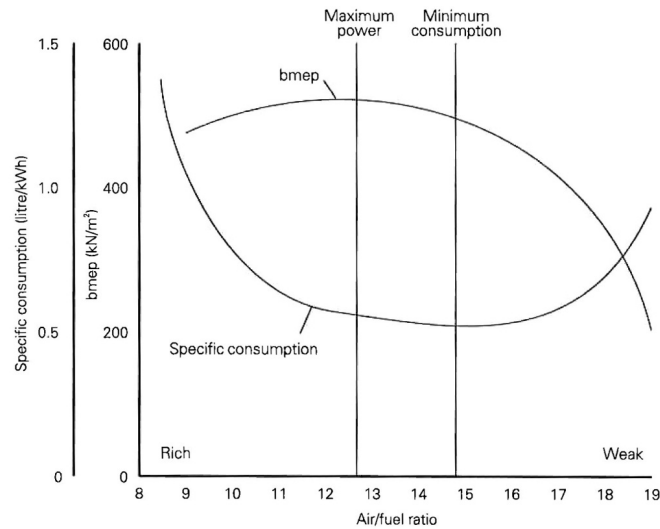
- To the left is a 'hook curve' plot
 - For this test the throttle and speed are kept constant while varying air/fuel ratio
- Point 'a' corresponds to the weakest mixture at which the engine will run
- Point 'b' corresponds to the best performance of the engine
 - Maximum thermal efficiency
- Point 'c' corresponds to the stoichiometric ratio
- Point 'd' corresponds to the maximum power
- Point 'e' corresponds to maximum mixture strength the engine will run



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[16]

Some IC Engine Background (3)

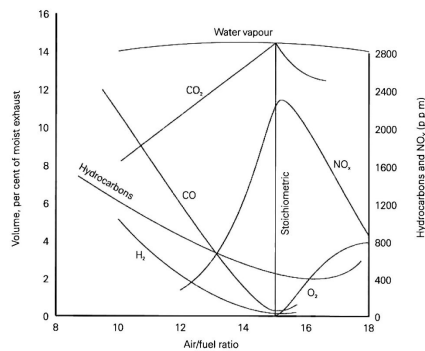


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[17]

Some IC Engine Background (4)

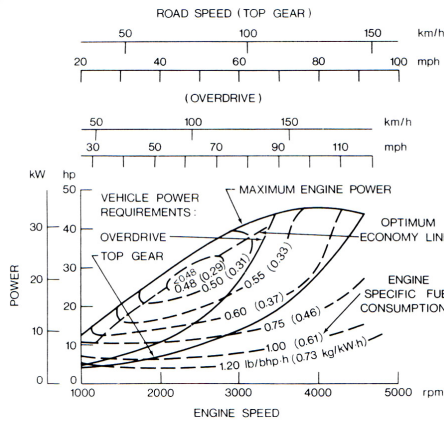
- The plot to the left relates ICE emissions to air/fuel ratio
- From the previous slide we saw
 - Max power at $\lambda \sim 0.9$ (A/F ~ 13)
 - At this point CO and hydrocarbon emissions are high
 - Max efficiency at $\lambda \sim 1.1$ (A/F ~ 15.6)
 - At this point NOx emissions are at a maximum



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[18]

SFC and MPG



- From the previous slide

$$\text{mpg} = \frac{\gamma V}{P_{\text{RL}} * \text{SFC}}$$

where,

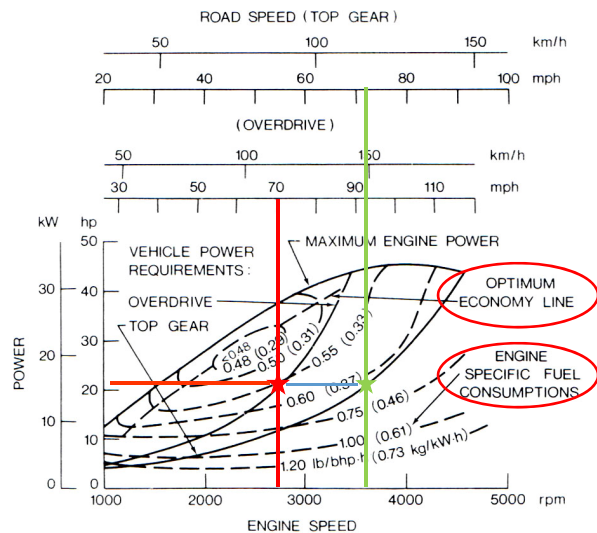
$$\gamma = \text{lb/gal}$$

- The road load power is independent of gearing and is only a function of the rolling and aerodynamic resistance for a given velocity
- Therefore, for a given velocity and therefore road load power the mpg is minimized by operating at the rpm that minimizes the SFC

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[19]

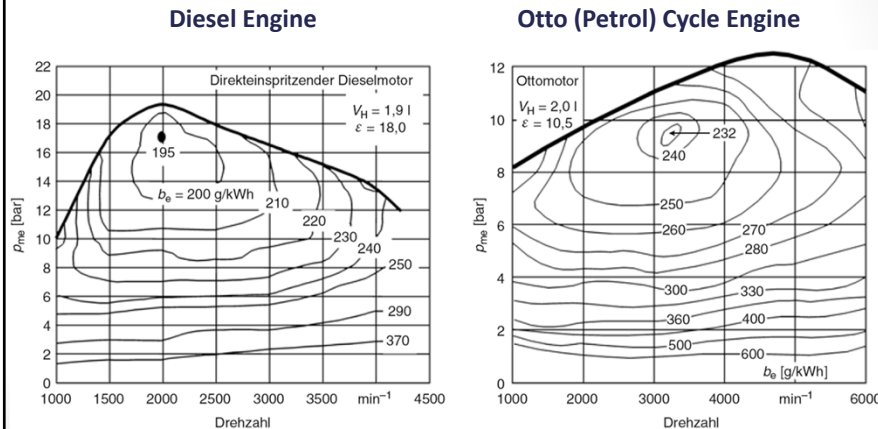
SFC and MPG (2)



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[20]

Characteristic Maps



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[21]

Transmission Efficiency and Ratios

- For an ideal transmission the input power is related to the output power by

$$P_o = \omega_o T_o = \frac{\omega_i}{\zeta} (\zeta T_i) = \omega_i T_i = P_i$$

- Where ξ is the gear ratio or speed ratio of input/output

$$\zeta = \frac{\omega_i}{\omega_o}$$

- As there are losses associated with any transmission we include an efficiency term

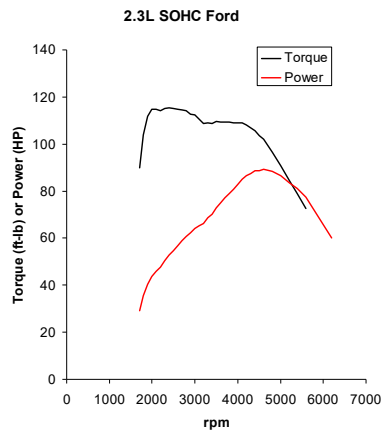
$$P_o = \eta P_i$$

- We must now choose our ratio(s) such that we meet our requirements for
 - Low speed operation/acceleration
 - Top speed operation/cruise efficiency
 - Gradability
 - Etc.

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[22]

Power Characteristics – Standard Data



- Generally, when we discuss power of the engine we mean the full throttle power vs. speed
 - However, this is only of partial use as it only bounds the problem for maximums/minimums
 - More useful will be a full characteristic map where we have the engine power and fuel usage over the full range of engine speeds, throttle and operating conditions
- The engine power must be corrected to the actual or standard conditions
 - Correcting to standard conditions

$$P = \frac{(B_a - B_v)}{B_o} \sqrt{\frac{T_o}{T}} P_o$$

where

$T_o = 288.5^\circ\text{K} = 60^\circ\text{F} = 15.5^\circ$

$B_o = 29.92 \text{ in Hg} = 76 \text{ cm Hg}$

P_o = uncorrected power

B_a = test cell barometric pressure at inlet

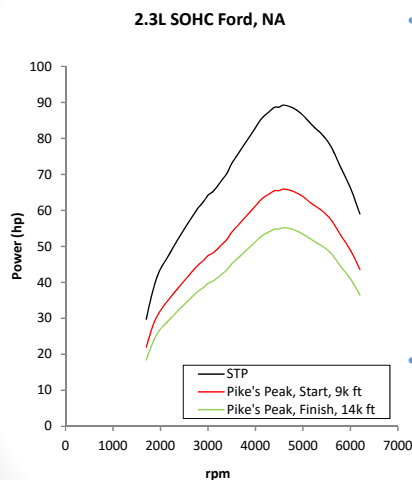
B_v = test cell vapor pressure (humidity)

T = test cell temperature

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(23)

Power Characteristics – Altitude



- Note that we can use the following equation and obtain local power from standard data

$$P = \frac{(B_a - B_v)}{B_o} \sqrt{\frac{T_o}{T}} P_o$$

where

$T_o = 288.5^\circ\text{K} = 60^\circ\text{F} = 15.5^\circ$

$B_o = 29.92 \text{ in Hg} = 76 \text{ cm Hg}$

P_o = uncorrected power

B_a = atm barometric pressure at inlet

B_v = atm vapor pressure (humidity)

T = atm temperature

- Consider Pikes Peak
 - A hill climb from 9000ft to 14000ft elevation
 - NA vs. Turbo?

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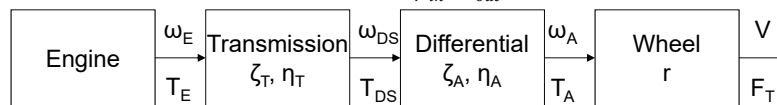
(24)

Transmission Ratios

- We can define

$$\eta = \text{efficiency} = \frac{P_{out}}{P_{in}} = \frac{2\pi n_{out} T_{out}}{2\pi n_{in} T_{in}} = \frac{n_{out} T_{out}}{n_{in} T_{in}}$$

$$\zeta = \text{gear ratio} = \frac{T_{out}}{\eta T_{in}} = \frac{\omega_{in}}{\omega_{out}}$$



where,

$$P_A = \omega_A T_A = \eta_A \eta_T P_E = \eta_A \eta_T \omega_E T_E$$

$$\omega_A = \frac{1}{\zeta_A \zeta_T} \omega_E \text{ and } T_A = \eta_A \eta_T \zeta_A \zeta_T T_E$$

and

$$V = r \omega_A = \frac{r}{\zeta_A \zeta_T} \omega_E \text{ and } F_T = \frac{1}{r} T_A = \frac{\eta_A \eta_T \zeta_A \zeta_T}{r} T_E$$

Trans/Diff Efficiency

Angular Velocity
and Torque
at Wheel
Centerline

Velocity and
Tractive Force
at Ground

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[25]

Transmission Ratio – First Gear

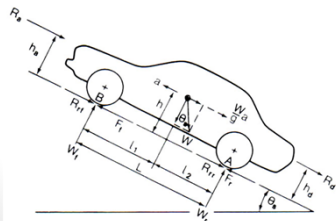
- First or bottom gear can also be chosen based on a variety of criteria, i.e.
 - Ease of pulling away from a stop
 - Maximum low speed acceleration
 - Gear to corner matching (generally not for first gear)
 - Gradability
- Gradability is also applicable to intermediate gears so let's consider it
 - Gradability can be limited by power or traction
 - The resistive load and required torque and therefore power can be determined as detailed previously
 - Traction limits will need to be developed

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[26]

Gradability – Traction Limited

- Assume
 - Constant low velocity
 - Air resistance neglected
 - Rotational inertia neglected
- Gradient angle, θ , limited by traction
 - Grade = $\tan \theta$
 - $F_w = m g \cos \theta$



- FWD

$$\tan \theta_f = \frac{\mu_{max} * \frac{l_2}{L} - F_w}{1 + \mu_{max} * \frac{h}{L}}$$

- RWD

$$\tan \theta_r = \frac{\mu_{max} * \frac{l_1}{L} - F_w}{1 - \mu_{max} * \frac{h}{L}}$$

- 4WD

$$\tan \theta_{AWD} = \mu_{max} - F_w$$

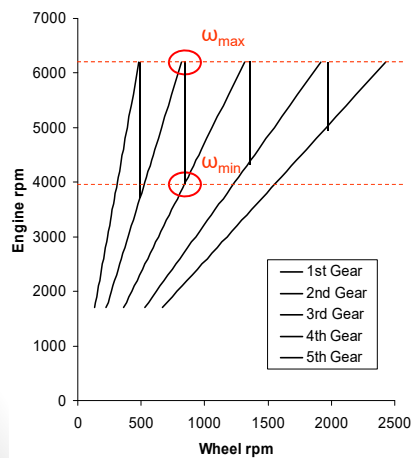
- If torque properly split

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[27]

Transmission Ratio – Intermediate Gears

2.3L SOHC Ford w/ 5-Speed Gearbox



- One way to specify the intermediate gears are to use a geometric progression between the 1st and final gear ratios

$$\omega_{max} = \xi_1 \omega_1 = \xi_2 \omega_2 = \xi_3 \omega_3 \dots$$

$$\omega_{min} = \xi_2 \omega_1 = \xi_3 \omega_2 = \xi_4 \omega_3 \dots$$

$$\frac{\omega_{min}}{\omega_{max}} = \frac{\xi_2}{\xi_1} = \frac{\xi_3}{\xi_2} = \frac{\xi_4}{\xi_3} \dots$$

$$\left(\frac{\omega_{min}}{\omega_{max}} \right)^3 = \frac{\xi_2}{\xi_1} \frac{\xi_3}{\xi_2} \frac{\xi_4}{\xi_3} = \frac{\xi_4}{\xi_1}$$

yielding,

$$K_G = \left(\frac{\omega_{min}}{\omega_{max}} \right) = \left(\frac{\xi_4}{\xi_1} \right)^{1/3}$$

$$= \left(\frac{\xi_n}{\xi_1} \right)^{\frac{1}{n-1}}$$

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[28]

Transmission Ratios – Tractive Effort

- Alternatively, we can calculate the tractive effort in each gear

$$F = \frac{1}{r} T_A = \frac{\eta_T \eta_A \xi_T \xi_A}{r} T_E$$

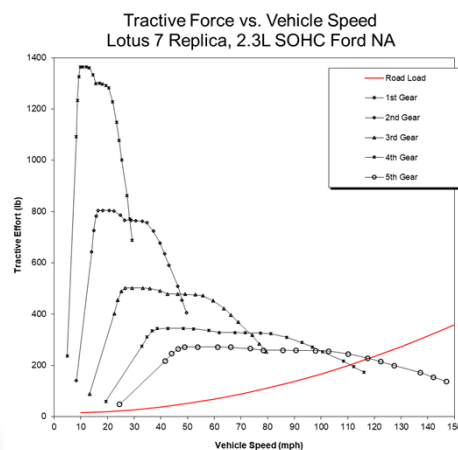
$$V = r \omega_A = \frac{r}{\xi_T \xi_A} \omega_E$$

- Then manipulate the overall ratios (differential and transmission) to achieve our desired objectives, i.e.
 - Top speed
 - Efficient cruising
 - Gear to corner matching
 - Acceleration
 - Etc.

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[29]

Tractive Effort – Road Load

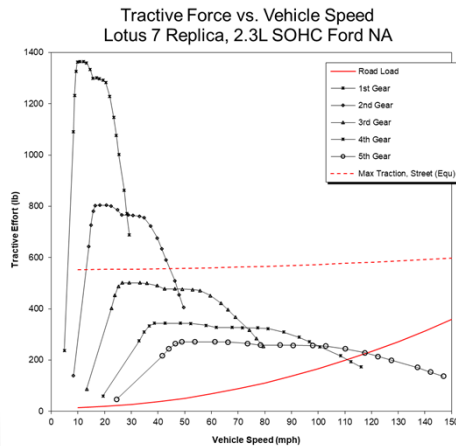


- For our 2.3L NA SOHC I4 Ford engine with a 5-speed gearbox we have the tractive effort curves for each gear shown to left
- Note the road load curve (rolling resistance plus aerodynamic drag in lbf)
 - The top speed and top gear ratios can be set with this method as well
 - Where are the shift points?
 - Why do the engine speed ranges cross for the higher gears but not for the lower gears?

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[30]

Tractive Effort – Traction Limits



- This driveline combination is intended for a Lotus 7 replica
 - 1300 lb weight target w/driver
 - c.g. height of 20 in.
 - Wheelbase of 90 in.
 - F/R 55/45
- We know for a RWD vehicle the maximum tractive force available

$$F_{T,max,RWD} = \mu_r W_r$$

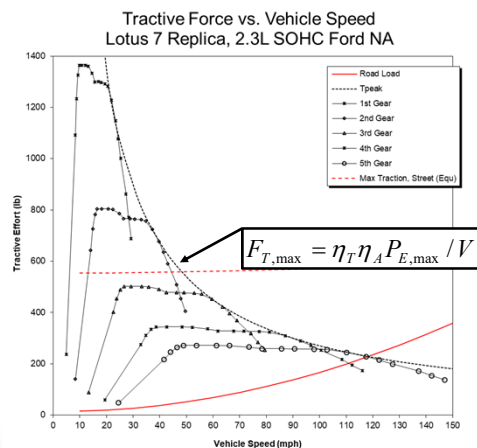
$$= \frac{\mu_r [(l_f - h\mu_r)(W - L_a) - M_a]}{(1 - \mu_r \frac{h}{L})}$$

- $f(\mu, m, h, L, \text{Lift, Drag, Moment})$
- Neglect aero forces for low speed

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[31]

Tractive Effort – Ideal Transmission?

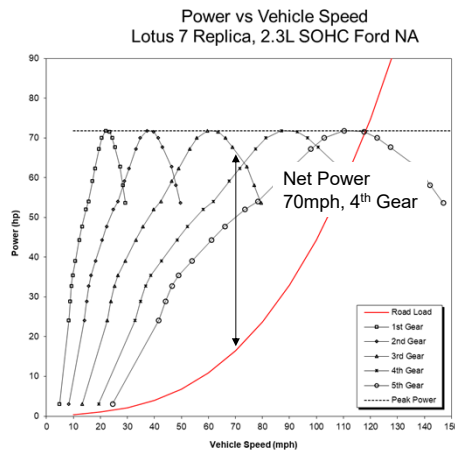


- Why would we want more gears?
- What is the maximum available tractive force from the engine?
- What type of transmission is ideal?
- What would the engine be doing with this transmission?
- Do you think the ideal transmission is necessarily the best choice? Why or why not?

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[32]

Net Power



- The net power is the difference between the output power and the road load power
- The net power is the power available for acceleration
- For instantaneous acceleration

$$F = ma$$

yielding,

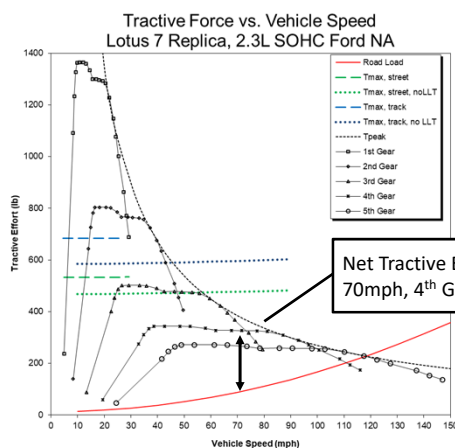
$$P_{\text{Net}} = \frac{Vma}{550}$$

$$a = \frac{550P_{\text{Net}}}{mV}$$

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[33]

Net Power or Tractive Effort



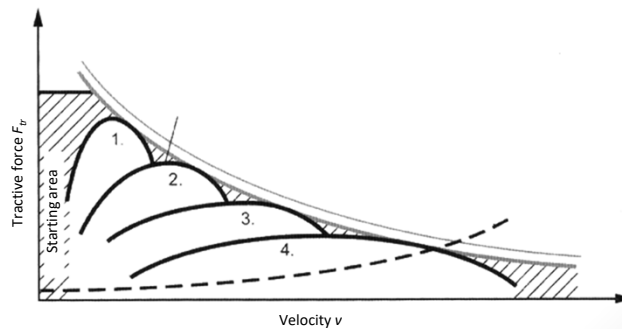
- Acceleration is proportional to net power and inversely proportional to momentum
- Acceleration decreases with speed
- This is one major reason why the upper gears are more closely spaced
- To determine the impact yourself consider an impulse-momentum case...

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[34]

What about the missing pieces?

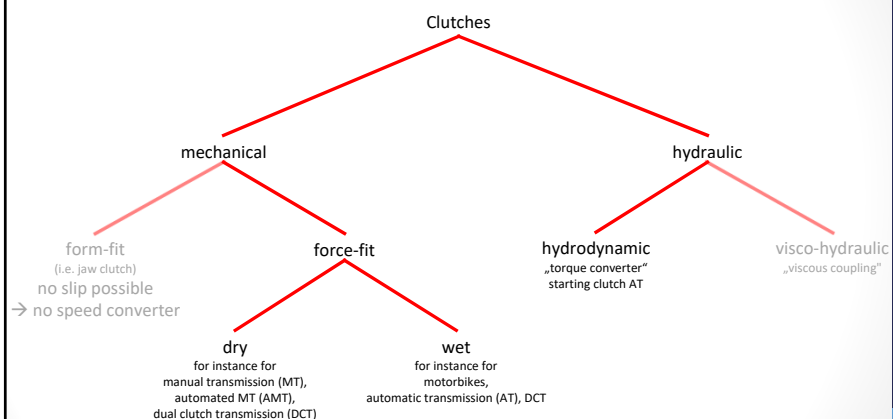
- The cross-hatched areas are not available for operation
 - Of particular interest is the starting area
 - In this area the engine would run too slow to operate or at minimum would not make a significant amount of torque
 - Some mechanism must be incorporated to convert torque across this gap
 - This is the function of a clutch



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[35]

Clutch Types

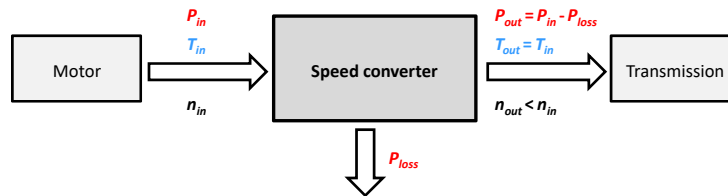


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[36]

Friction Mechanical Clutch

- Difference in speed
- Torque constant
- Power loss



- Efficiency

$$\eta = \frac{P_{out}}{P_{in}} = \frac{2\pi n_{out} T_{out}}{2\pi n_{in} T_{in}} = \frac{n_{out}}{n_{in}}$$

- Slip

$$s = \frac{n_{in} - n_{out}}{n_{in}} = 1 - \frac{n_{out}}{n_{in}} = 1 - \eta$$

- Power loss

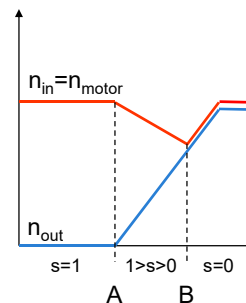
$$P_{loss} = P_{in} - P_{out} = (1 - \eta)P_{in} = sP_{in}$$

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[37]

Mechanical Clutch Engagement

- Clutch engages at A
 - Rotational speed of clutch (output) increases and engine (input) decreases
 - Therefore the rotational (mass) inertia of the augments the input torque
- At B the clutch is fully engaged (no slip, $s=0$)
- To avoid a jerk at the end of engagement the sliding and static coefficients of friction for the clutch should be approximately the same



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[38]

Hydrodynamic Converter

- Starting clutch for automatics and CVTs
- Provides torque transmission with slip only
- Can increase torque but up to 3x
- Consists of an impeller, stator and turbine with an oil working fluid

- The speed differential between impeller, n_p , and turbine, n_T is

$$v = \frac{n_T}{n_p}$$

- The converter torque ratio is the ratio of the turbine and impeller torques

$$\mu = \frac{T_T}{T_p}$$

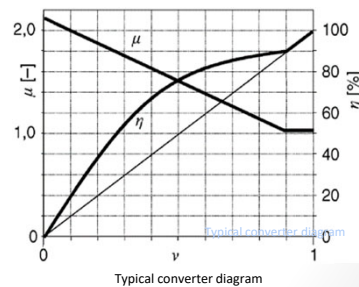
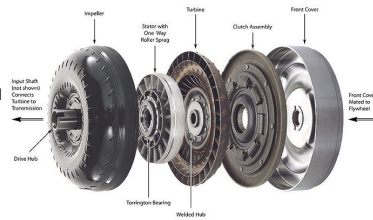
- The efficiency is defined

$$\eta = \frac{T_T * n_T}{T_p * n_p} = \mu * v$$

- The power factor, C, for the calculation of transferable torque by the impeller, TP depends on the oil and converter geometry

- A representative converter diagram is shown to the right

- Torque augmentation is initially 2.1 and falls to 1.0 at $v=0.9$
 - At $v=0.9$ a lockup clutch engages for increased efficiency



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[39]

Clutch Comparison

Mechanical

- Dry
- Advantage
 - Simple, low-cost
 - Minimal drag when disengaged
 - Independent system
- Disadvantage
 - Limited torque capacity
 - Wear
 - Low heat capacity (heavy)
 - No continuous slip possible
 - Must be kept clean
- Wet (multi-plate)
- Advantage
 - High torque capacity
 - Low release force
 - Cooling by oil flow
 - Free of wear (essentially)
 - Permanent slip possible
- Disadvantage
 - Drag torque when disengaged
 - Cost
 - Oil pollution from wear
 - length

Hydrodynamic

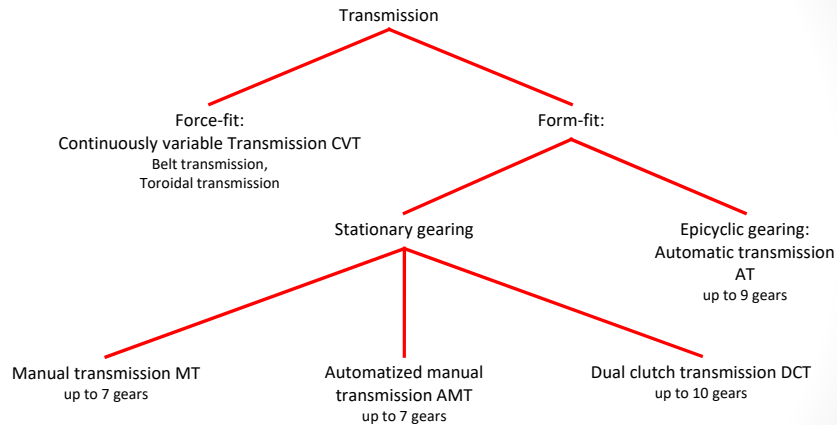
- Advantage
 - self-regulating
 - high thermal stability
 - free of wear
 - continuous slip operation possible
 - independent system
- Disadvantage
 - $T_r = 0$ for $\Delta n = 0$: torque transfer only with slip
 - $s > 0 \rightarrow \eta < 1$: slip causes losses
 - for $\eta = 1$ additional effort required: converter interlock clutch
 - limited rotational speed capacity
 - no control possible
 - high drag torque at idle $\rightarrow$ only appropriate for power shift transmissions

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[40]

Other types of clutch: visco-hydraulic and jaw (dog) clutch

Transmission Types



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(41)

Time to Speed

- Time to speed is an important measure of average acceleration
 - 0-60mph, 0-100mph times are common performance measures of performance for the consumer
 - 30-60mph is important for merging with freeway traffic
 - 60-80mph is important for passing at freeway speeds

- We can define the acceleration

$$F_T = ma$$

where,

F_T = tractive force

m = effective mass

a = acceleration

and

$$F_T = m \frac{dV}{dt}$$

$$dt = \frac{mdV}{F_T}$$

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(42)

Time to Speed (2)

- We can determine the Time to Speed (TTS) as a function of
 - Tractive force as a function of velocity

$$TTS = \int_{V_1}^{V_2} \frac{m dV}{F_T(V)}$$

- Net power as a function of velocity

$$TTS = \int_{V_1}^{V_2} \frac{m V dV}{P_{net}(V)}$$

- These integral relations can be solved in Matlab or similar
- Alternatively, we can do this numerically in Excel

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[43]

Time to Speed – Numerical Calculation

1. We will use the relations

$$F_T = m \frac{dV}{dt}$$

$$dt = \frac{m dV}{F_T}$$

1. Obtain the torque vs. engine speed data for your vehicle
2. Calculate the road speed in each gear
3. Determine the shift points for each gear
 - Determine points in sequential gears where the speeds match
 - From the tractive effort or horsepower vs. road speed curves
4. Calculate the road load power
5. Calculate, P_{Net}

$$P_{Net} = P_e - P_{RL}$$

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[44]

Time to Speed – Numerical Calculation

8. Calculate the possible tractive forces
 - Tractive effort due to engine torque
 - Tractive effort limited by maximum traction
9. Select the lower of the two possible tractive forces and calculate

$$\frac{dt}{dV} = \frac{m}{F_T}$$

10. Calculate Δt

$$\Delta t_i = \frac{\left[\left(\frac{dt}{dV} \right)_{i-1} + \left(\frac{dt}{dV} \right)_i \right] * [(V)_i - (V)_{i-1}]}{2}$$

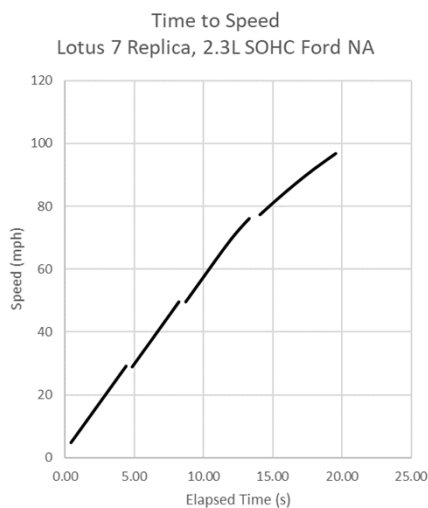
8. Calculate TTS by summing Δt

$$\sum_{i=V_i}^{V_f} \Delta t_i$$

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[45]

TTS – Lotus 7 Replica, 2.3L Ford NA



- For the Lotus 7 replica
 - 2.3L SOHC Ford NA
 - 5-speed transmission
 - 1300 lb weight target w/driver
 - c.g. height of 20 in.
 - Wheelbase of 90 in.
 - F/R 55/45
- And the following driver skills
 - Maintain max traction in the slip range
 - 0.5s shift times
- TTS, street tires
 - 0-60mph: 10.2s (3rd gear)
 - 0-100mph: 21.2s (4th gear)
 - Traction limited up to 63mph in 3rd gear at ~4500rpm
- TTS, track-day tires ('sport box')
 - 0-60mph: 8.9s (2nd gear)
 - 0-100mph: 18.6s (4th gear)
 - Traction limited up to 50mph in 2nd gear at 5000rpm but more F_T up until then...

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[46]